

Estimating dispersion

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EDMs and GLMs

We say that a distribution belongs to the **exponential dispersion model** family if the distribution's probability function can be written in the form

$$f(y; \theta, \phi) = a(y, \phi) \exp \left\{ \frac{y\theta - \kappa(\theta)}{\phi} \right\}$$

$$f(y; \mu, \phi) = b(y, \phi) \exp \left\{ -\frac{d(y, \mu)}{2\phi} \right\}$$

For iid observations $(\mathbf{x}_1, Y_1), \dots, (\mathbf{x}_n, Y_n)$, a **generalized linear model** (GLM) consists of two components:

$$Y_i | \mathbf{x}_i \sim \text{EDM}(\mu_i, \phi)$$

$$g(\mu_i) = \mathbf{x}_i^T \boldsymbol{\beta}$$

Estimating β

$$Y_i | \mathbf{x}_i \sim \text{EDM}(\mu_i, \phi)$$
$$g(\mu_i) = \mathbf{x}_i^T \beta$$

Newton's method:

- ▶ $U(\beta) = \frac{1}{\phi} \sum_{i=1}^n (Y_i - \mu_i) \mathbf{x}_i = \frac{1}{\phi} \mathbf{X}^T (\mathbf{y} - \boldsymbol{\mu})$
- ▶ $\mathbf{H}(\beta) = -\frac{1}{\phi} \sum_{i=1}^n V(\mu_i) \mathbf{x}_i \mathbf{x}_i^T$
- ▶ $\mathcal{I}_1(\beta) = \frac{1}{\phi} \mathbb{E}[V(\mu_i) \mathbf{x}_i \mathbf{x}_i^T]$

Estimating ϕ

Dispersion model form:

$$f(y; \mu, \phi) = b(y, \phi) \exp \left\{ -\frac{d(y, \mu)}{2\phi} \right\}$$

Approximating $b(y, \phi)$

Saddlepoint approximation

MLE for ϕ and the mean deviance estimator

$$f(y; \mu, \phi) = b(y, \phi) \exp \left\{ -\frac{d(y, \mu)}{2\phi} \right\} \approx \frac{1}{\sqrt{2\pi\phi V(y)}} \exp \left\{ -\frac{d(y, \mu)}{2\phi} \right\}$$

Pearson estimator

Activity

Work on the activity (course website).